

Active Brownian particles with mean-field interaction

Singular interactions and multiple invariant measures

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Active Brownian particles

- 2D Active Brownian particles

- Particle i follows
$$\begin{cases} dX_t^i &= \text{Pev}(\theta_t^i)dt + \sqrt{2\sigma_x}dW_t^i \\ d\theta_t^i &= \sqrt{2}dB_t^i \end{cases}$$